

# On the Dynamics of Gravity Oscillations in a Shallow Water Model, with Applications to Normal Mode Initialization

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**Abstract:** A spectral, hemispheric shallow water model with a mean height equal to 8800 m is considered. Free normal mode solutions are determined and compared to gravity oscillations observed in the nonlinear numerical model. A very good agreement is found with the normal modes and their frequencies, however, a discrepancy is found between the observed amplitudes and those determined by the free normal mode solutions. This discrepancy is removed by taking into account the nonlinear interactions between the normal modes. A new normal mode filtering procedure, intended for the **elimination of unwanted gravity oscillations is presented**. The procedure imply that the time derivative of the normal mode coefficients for unwanted gravity modes are set equal to zero initially. Experiments show that by this procedure all gravity oscillations, including the most large scale components, can be eliminated. The results of the new procedure differ from the free normal mode filtering procedure proposed by DICKINSON and WILLIAMSON (1972), especially, for the relatively large-scale (low-frequency) gravity modes. The application of the filtering procedure to initialization of baroclinic models is finally discussed.

## 1. Introduction

The dynamics of gravity oscillations in numerical models, designed for the simulation of large-scale atmospheric processes, seems to be of interest primary for two reasons. Firstly they may be generated in the numerical models with unrealistic large amplitudes due to insertions of erroneous data or due to errors in the numerical model. Secondly being those oscillations in primitive equation models with highest frequencies they are of importance for the stability of the finite time difference schemes used in the models. The present study is aimed at the former of these areas of interest.

We shall consider gravity oscillations in a shallow water model, corresponding to large-scale external gravity oscillations in the atmosphere. The general opinion is that such large-scale, high-frequency oscillations are of very small amplitude in the atmosphere. Thus, these gravity oscillations themselves are probably of no importance in the simulation of large scale atmospheric processes; and if present with large amplitudes in a numerical model they are signs of errors. We must, nevertheless, understand the dynamics of these oscillations in order to be able to isolate the reasons for their presence and to correct the errors. We will consider, especially, the characteristics of the initial fields which give rise to gravity oscillations in the subsequent integration period, and methods by which such oscillations may be eliminated.

Methods aimed at the specification of suitable balanced initial fields, which give rise to no spurious gravity oscillations, are known as methods of initialization. Several such methods have been used in numerical prediction, and it would lead too far just to mention the most important ones; the reader is referred to BENGTTSSON (1975) for a review and a discussion of the methods. Here we shall only mention that no method have been capable of completely eliminating gravity oscillations and often the methods have undesirable secondary effects. Difficulties have been encountered especially in global models and here in particular problems have arisen in connection with the elimination of the most large scale gravity modes.

Recently a relatively new method, building upon normal mode solutions to linearized systems of equations, has gained interest. The method was introduced by FLATTERY (1970). In his Hough function analysis, the atmospheric data are expanded in terms of the ROSSBY modes of the free oscillations in the shallow water equations, linearized around a basic state at rest. At all levels the same ROSSBY modes are used, namely those corresponding to a certain mean height of the shallow water model. The use of only the ROSSBY modes imply that no gravity oscillations would appear in an exact integration of the linear shallow water equations, when using a mean height equal to that chosen for the analysis. When used as initial data to a nonlinear baroclinic model oscillations of the gravity wave type will, however, still be generated.

DICKINSON and WILLIAMSON (1972) propose a normal mode filtering method to be designed especially for a specific numerical model. A general approach to determine the normal modes of a numerical model was presented by these authors and illustrated by application to a global finite difference two-layer model. They propose to expand the model data in terms of the complete set of normal modes, and then to reduce or set equal to zero the expansion coefficients of unwanted computational and gravity modes.

WILLIAMSON (1976) has recently applied this procedure to a shallow water grid point model. Using a basic state at rest, the expansion coefficients of the computational and gravity modes were set equal to zero initially. This procedure did reduce the gravity oscillations in the nonlinear integrations, however, some oscillations apparently of the gravity type remained.

The present study may be considered as an extension of the works with normal mode filtering referred to above. The main difference is, that we take into account the initial forcing of the gravity oscillations by the nonlinear terms in the model equations. Furthermore the method is applied to a spectral model.

The model considered is a spectral shallow water model, similar to that described by BOURKE (1972). A hemisphere version is used with a mean height of the upper surface,  $\bar{H}$ , equal to 8800 m, which corresponds approximately to the equivalent height for external modes in baroclinic models. In all experiments were used rhomboidal truncation at wave number 17 (that is,  $M = J = 17$  in Equation (1)) and the leap frog time integration scheme, with a time step  $\Delta t = 0,125$  hours. In order to suppress time computational modes, a smooth start was used, that is, each integration was started with a forward time step of  $\Delta t/4$  followed by two centered time steps of  $\Delta t/2$  and  $\Delta t$ , respectively.

In Section 2 the equations of the model are transformed to normal mode form, free normal mode solutions are found and the filtering method, based on these solutions, is presented. In Section 3 the free normal mode solutions are compared to the observed gravity oscillations in the model. In Section 4 solutions taking into account the nonlinear interactions are presented and a new filtering method based on these solutions is presented. In Section 5 results of experiments with the filtering methods are presented and in Section 6 the applications of the new filtering method are discussed.

## 2. Elimination of Gravity Oscillations Based on Free Normal Mode Solutions

The variables of the shallow water model, the velocity field,  $\vec{V}$ , and the height of the free surface,  $h$ , may be written as

$$\vec{V}(\lambda, \varphi, t) = U(\varphi) \vec{i} + \vec{V}'(\lambda, \varphi, t)$$

$$h(\lambda, \varphi, t) = H(\varphi) + z(\lambda, \varphi, t)$$

where  $\vec{V}' = u\vec{i} + v\vec{j} = \vec{k} \times \nabla \psi + \nabla \alpha$  and  $z$  are the deviations from a stationary, zonal basic state determined by  $U(\varphi)$  and  $H(\varphi)$ . As usual  $\lambda$ ,  $\varphi$  and  $t$  are the longitude, latitude and time, and  $\vec{i}$ ,  $\vec{j}$  and  $\vec{k}$  are the unit vectors towards the east, the north and in the vertical direction, respectively. The basic stream may be specified by a given angular velocity  $\omega(\varphi)$  as

$$U(\varphi) = a \omega(\varphi) \cos \varphi$$



where  $a$  is the radius of the earth. The deviation from the mean height  $\bar{H}$  of the basic height field  $H(\varphi)$  is then determined from the nonlinear balance equation

$$\frac{1}{a^2 \cos \varphi} \frac{\partial}{\partial \varphi} (U^2 \sin \varphi) + U \frac{2\Omega}{a} \cos \varphi + \frac{2\Omega}{a} \tan \varphi \frac{\partial}{\partial \varphi} (U \cos \varphi) = -g \nabla^2 H$$

where  $g$  is gravity,  $\Omega$  is the angular velocity of the earth and  $\nabla^2$  is the surface spherical Laplacian operator. In the case of a solid rotating basic state, that is,  $\omega$  equal to a constant, we get

$$H(\varphi) = \bar{H} + \frac{\Omega a^2}{2g} \left( \omega + \frac{\omega^2}{2\Omega} \right) \left( \cos 2\varphi - \frac{1}{3} \right)$$

which reduces to  $H = \bar{H}$  in the case  $\omega = 0$ , a basic state at rest.

The equations for the shallow water model may now be written

$$\frac{\partial \xi}{\partial t} = -\omega \frac{\partial \xi}{\partial \lambda} - \beta' v - f' \delta + Q_\psi$$

$$\frac{\partial \delta}{\partial t} = -\omega \frac{\partial \delta}{\partial \lambda} - \beta' u + f' \xi - g \nabla^2 z + Q_\alpha$$

$$\frac{\partial z}{\partial t} = -\omega \frac{\partial z}{\partial \lambda} - v \frac{\partial H}{a \partial \varphi} - H \delta + Q_z$$

where  $\xi = \nabla^2 \psi$ ,  $\delta = \nabla^2 \alpha$ ,  $f' = 2\Omega \sin \varphi - \frac{\partial U \cos \varphi}{a \cos \varphi \partial \varphi}$  and  $\beta' = \frac{\partial f'}{a \partial \varphi}$

On the right hand side the linear terms have been written explicitly, whereas the nonlinear terms are represented by the  $Q$  terms:

$$Q_\psi = -\nabla \cdot \xi \vec{V}'$$

$$Q_\alpha = \vec{k} \cdot \nabla \times \xi \vec{V}' - \nabla^2 \frac{\vec{V}' \cdot \vec{V}'}{2}$$

$$Q_z = -\nabla \cdot z \vec{V}'$$

Introducing the following spectral representation for the variables over a hemisphere

$$\begin{aligned} \psi(\lambda, \varphi, t) &= \frac{2\Omega^2 a^3}{10^4 \sqrt{gH}} \sum_{m=-M}^M \sum_{n=|m|+1}^{|m|+J} (2) \frac{\psi_{m,n}(t)}{\sqrt{n(n+1)}} P_{m,n}(\varphi) e^{im\lambda} \\ \alpha(\lambda, \varphi, t) &= \frac{2\Omega^2 a^3}{10^4 \sqrt{gH}} \sum_{m=-M}^M \sum_{n=|m|}^{|m|+J-1} (2) \frac{\alpha_{m,n}(t)}{\sqrt{n(n+1)}} P_{m,n}(\varphi) e^{im\lambda} \\ z(\lambda, \varphi, t) &= \frac{2\Omega^2 a^2}{10^4 g} \sum_{m=-M}^M \sum_{n=|m|}^{|m|+J-1} (2) \eta_{m,n}(t) P_{m,n}(\varphi) e^{im\lambda} \end{aligned} \quad (1)$$

where  $P_{m,n}(\varphi)$  are normalized, associated Legendre functions of order  $m$  and degree  $n$ , the equations may be transformed to the spectral form

$$\frac{d}{dt} X_{m|} = i A_{m|} X_{m|} + Q_{m|} \quad (2)$$

where  $a$  is the radius of the earth. The deviation from the mean height  $\bar{H}$  of the basic height field  $H(\varphi)$  is then determined from the nonlinear balance equation

$$\frac{1}{a^2 \cos \varphi} \frac{\partial}{\partial \varphi} (U^2 \sin \varphi) + U \frac{2\Omega}{a} \cos \varphi + \frac{2\Omega}{a} \tan \varphi \frac{\partial}{\partial \varphi} (U \cos \varphi) = -g \nabla^2 H$$

where  $g$  is gravity,  $\Omega$  is the angular velocity of the earth and  $\nabla^2$  is the surface spherical Laplacian operator. In the case of a solid rotating basic state, that is,  $\omega$  equal to a constant, we get

$$H(\varphi) = \bar{H} + \frac{\Omega a^2}{2g} \left( \omega + \frac{\omega^2}{2\Omega} \right) \left( \cos 2\varphi - \frac{1}{3} \right)$$

which reduces to  $H = \bar{H}$  in the case  $\omega = 0$ , a basic state at rest.

The equations for the shallow water model may now be written

$$\frac{\partial \zeta}{\partial t} = -\omega \frac{\partial \zeta}{\partial \lambda} - \beta' v - f' \delta + Q_\psi$$

$$\frac{\partial \delta}{\partial t} = -\omega \frac{\partial \delta}{\partial \lambda} - \beta' u + f' \zeta - g \nabla^2 z + Q_\alpha$$

$$\frac{\partial z}{\partial t} = -\omega \frac{\partial z}{\partial \lambda} - v \frac{\partial H}{a \partial \varphi} - H \delta + Q_z$$

where  $\zeta = \nabla^2 \psi$ ,  $\delta = \nabla^2 \alpha$ ,  $f' = 2\Omega \sin \varphi - \frac{\partial U \cos \varphi}{a \cos \varphi \partial \varphi}$  and  $\beta' = \frac{\partial f'}{a \partial \varphi}$

On the right hand side the linear terms have been written explicitly, whereas the nonlinear terms are represented by the  $Q$  terms:

$$Q_\psi = -\nabla \cdot \zeta \vec{V}'$$

$$Q_\alpha = \vec{k} \cdot \nabla \times \zeta \vec{V}' - \nabla^2 \frac{\vec{V}' \cdot \vec{V}'}{2}$$

$$Q_z = -\nabla \cdot z \vec{V}'$$

Introducing the following spectral representation for the variables over a hemisphere

$$\begin{aligned} \psi(\lambda, \varphi, t) &= \frac{2\Omega^2 a^3}{10^4 \sqrt{gH}} \sum_{m=-M}^M \sum_{n=|m|+1}^{|m|+J} (2) \frac{\psi_{m,n}(t)}{\sqrt{n(n+1)}} P_{m,n}(\varphi) e^{im\lambda} \\ \alpha(\lambda, \varphi, t) &= \frac{2\Omega^2 a^3}{10^4 \sqrt{gH}} \sum_{m=-M}^M \sum_{n=|m|}^{|m|+J-1} (2) \frac{\alpha_{m,n}(t)}{\sqrt{n(n+1)}} P_{m,n}(\varphi) e^{im\lambda} \\ z(\lambda, \varphi, t) &= \frac{2\Omega^2 a^2}{10^4 g} \sum_{m=-M}^M \sum_{n=|m|}^{|m|+J-1} (2) \eta_{m,n}(t) P_{m,n}(\varphi) e^{im\lambda} \end{aligned} \quad (1)$$

where  $P_{m,n}(\varphi)$  are normalized, associated Legendre functions of order  $m$  and degree  $n$ , the equations may be transformed to the spectral form

$$\frac{d}{dt} X_{m|} = i \underline{A_m} X_{m|} + Q_{m|} \quad (2)$$

Here the column vector  $X_{m|}$  for a certain zonal wave number  $m$  has  $N = 1.5(J + 1)$  components,  $x_{m|}^j$ , determined by the complex spherical harmonic coefficients as follows

$$x_{m|}^j = \begin{cases} \eta_{m, |m|} + \frac{2}{3}(j-1) & \text{for } j = 1, 4, 7, \dots, N-2 \\ -i\alpha_{m, |m|} + \frac{2}{3}(j-2) & \text{for } j = 2, 5, 8, \dots, N-1 \\ \psi_{m, |m|} + \frac{2}{3}j-1 & \text{for } j = 3, 6, 9, \dots, N \end{cases}$$

and  $Q_{m|}$  has components  $q_{m|}^j$  equal to the nonlinear contributions to the time derivatives of the corresponding components in  $X_{m|}$ . As  $\psi$ ,  $\alpha$ , and  $z$  are real, it follows that we need to consider only those Equations (2) for which  $m \geq 0$ . In each Equation (2) all the linear terms are included in the first term on the right hand side, and here  $\underline{A}_m$  is a  $N \times N$  matrix with real components.

In the case of a basic state at rest, that is, when  $\omega$  has been chosen equal to zero,  $\underline{A}_m$  becomes symmetric. This is a consequence of the choice of the normalizing factors in (1), which have been chosen so that

$$E' = \frac{1}{2\pi a^2 \bar{H}} \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \left( \frac{1}{2} \vec{V}' \cdot \vec{V}' \bar{H} + \frac{1}{2} z^2 g \right) a^2 \cos \varphi d\varphi d\lambda$$

$$= \frac{2\Omega^4 a^4}{10^8 g \bar{H}} \sum_{m=-M}^M \sum_{n=|m|}^J (\psi_{m, n} \psi_{m, n}^* + \alpha_{m, n} \alpha_{m, n}^* + \eta_{m, n} \eta_{m, n}^*)$$

For the linear model,  $\omega = 0$ ,  $Q_{m|} = 0$ ,  $E'$  is a convenient measure of the total energy per unit mass of the deviation fields, and as  $\frac{d}{dt} E' = 0$  it follows that  $\underline{A}_m$  is symmetric.

Solving for each  $m \geq 0$  the eigenvalue problem

$$(\underline{A}_m - \sigma_m \underline{I}) X_{m|} = 0$$

we get  $N$  eigenvalues  $\sigma_m^k$  ( $k = 1, 2, \dots, N$ ) and the corresponding normalized eigenvectors  $X_{m|}^k$ , which become orthogonal in case of a basic state at rest.

Defining the linear substitution

$$X_{m|} = \underline{T}_m Y_{m|} \quad (3)$$

where the column vectors in the matrix  $\underline{T}_m$  are the eigenvectors  $X_{m|}^k$  and applying this substitution to Equation (2) gives

$$\frac{d}{dt} Y_{m|} = i \underline{\Gamma}_m Y_{m|} + R_{m|} \quad (4)$$

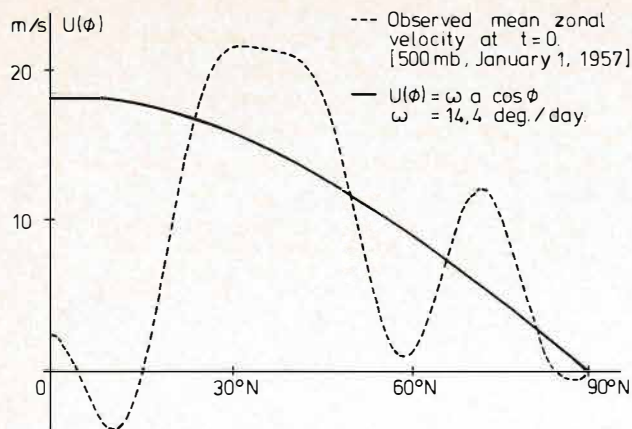
where  $\underline{\Gamma}_m$  is diagonal with the eigenvalues  $\sigma_m^k$  in the diagonal, and where

$$R_{m|} = \underline{T}_m^{-1} Q_{m|} \quad (5)$$

For one component in Equation (4) we have

$$\frac{d}{dt} y_m^k = i \sigma_m^k y_m^k + r_m^k \quad (6)$$





● Figure 1

Mean zonal velocity profiles defining the basic states with  $\omega \neq 0$ , used in Figure 2 b.

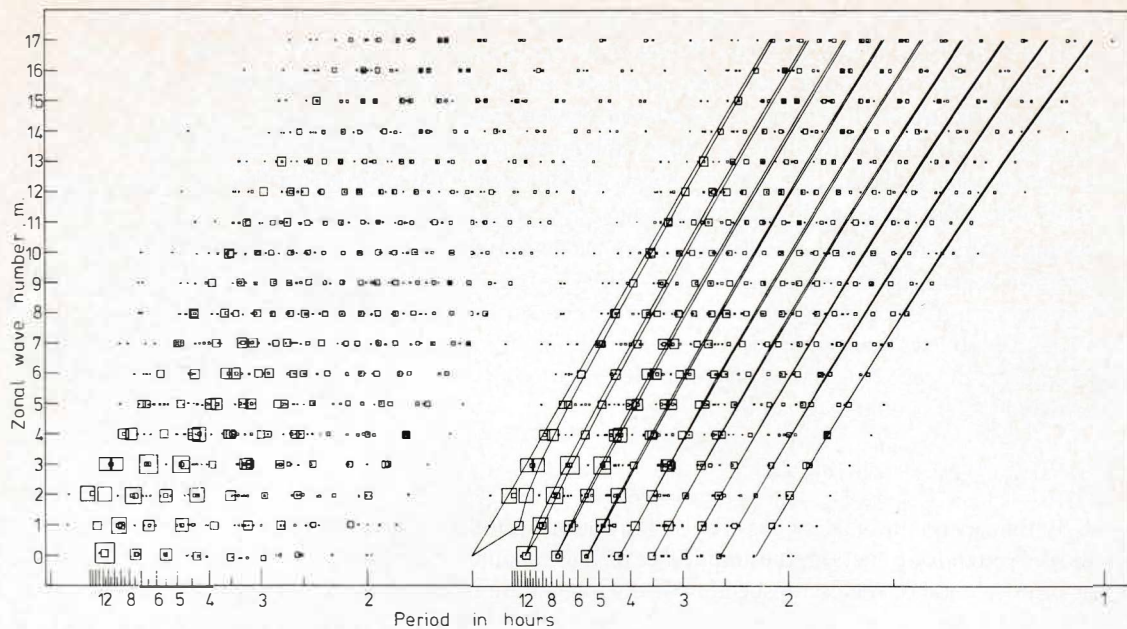
conclusion can be drawn from the experiments by WILLIAMSON (1976) with a shallow water gridpoint model. As seen from his Figure 1 changes over a few hours of the height of the free surface are reduced approximately from 250 m to 50 m after the filtering.

### 3. Gravity Oscillations in the Nonlinear Model

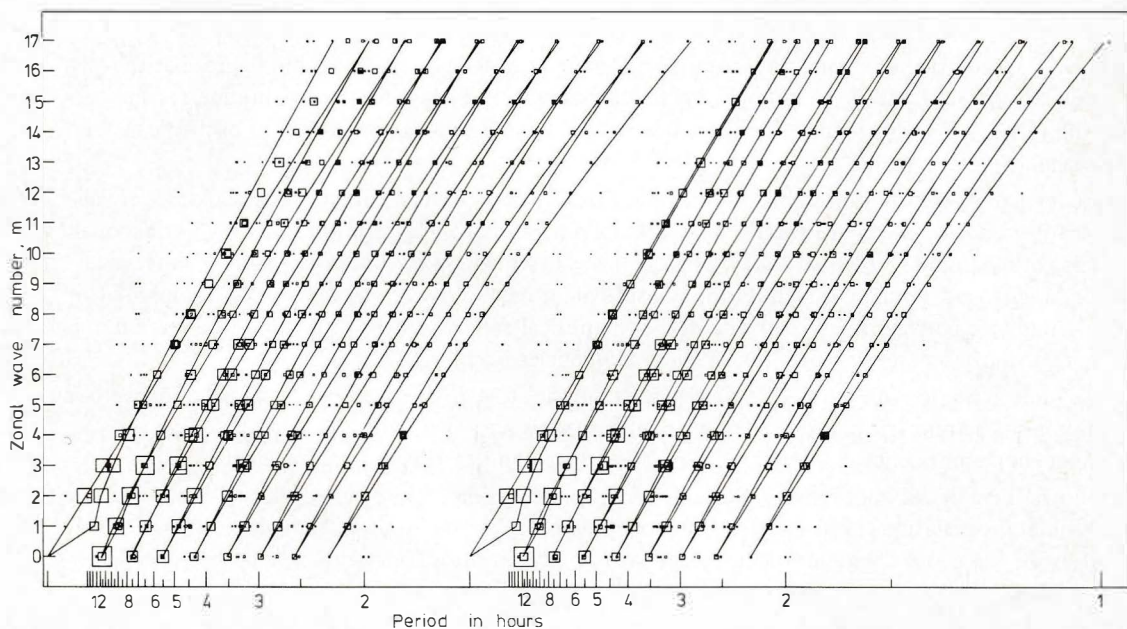
The relevance of the filtering procedure described in the preceding section depends upon to what extent the gravity oscillations in the nonlinear numerical model correspond with those determined by the linear theory. In order to investigate this question we have made spectral analysis of the time series obtained for each spherical harmonic coefficient during a 6 days forecast period. The initial velocity field was chosen nondivergent, determined from a preliminary height field obtained by a spherical harmonic analysis of the 500 mb height field from 1st January 1957 using the linear balance equation as described by ELIASEN and MACHENHAUER (1965). Using the velocity field obtained by this procedure the initial height field was then determined from the "reverse" linear balance equation. Thus the initial fields satisfy the linear balance equation exactly.

The time series were analysed by the Maximum Entropy Method (MEM) as described by JOHNSEN and ANDERSEN (1976). This method for spectral analysis is a relatively new method, which in general is superior to more conventional methods of spectral estimations. For each time series all data obtained during the 6 days forecast period were used, that is, each time series was composed of 1153 values. The so-called prediction filter length, which must be specified in the MEM, was chosen equal to 100.

It is not possible here to bring the results of the time series analysis for all spherical harmonic coefficients. We have chosen to show only some results for the velocity potential components and to present these in a rather compact form, which makes a comparison with the frequencies to be expected from linear theory easy. On the left hand side of Figure 2a oscillations found in the time series of these components are indicated by boxes. The area of a box is proportional to the amplitude of the oscillation, the frequency is given by the abscisse and the zonal wave number by the ordinate of the center of the box. (A linear scale in frequency is used at the abscisse axis, however, instead of frequencies, the values at the axis are the corresponding periods in hours). Thus, the oscillations found for all velocity potential components with the same zonal wave number are indicated by boxes on the same "horizontal" line. Only oscillations with amplitude above a certain lower limit, with periods below 40 hours for  $m \leq 3$  and with periods below 12 hours for  $m > 3$  are indicated in the figure.



● Figure 2a. Results of time series analysis for spherical harmonic components of velocity potential, in the right hand side of the figure with superimposed lines indicating theoretical frequencies,  $|(\sigma_m^k)_c|$ , computed using a basic state at rest.



● Figure 2 b. As figure 2 a, with lines indicating theoretical frequencies, in the left hand side using the solid rotation and in the right hand side the mean zonal velocity at  $t = 0$ , shown in Figure 1.



Consider for instance the boxes centered around the point in the diagram given by the zonal wave number  $m = 0$  and the period 12 hours. We see three boxes centered closely to this point: A big one, a smaller one inside the big one, and a very small one on the left side of the latter. These three boxes show that oscillations with periods close to 12 hours were found for three velocity potential components with  $m = 0$ . It is of course not possible to see from the figure for which components these oscillations were found. We may, however, give the additional information that the three boxes correspond to oscillations in the components  $\alpha_{0,2}$ ,  $\alpha_{0,4}$ , and  $\alpha_{0,6}$  arranged after decreasing amplitude.

According to the linear theory described in the preceding section (cf. Equation (7) and (9)) the time dependence of a spherical harmonic coefficient should be given as a superposition of harmonic oscillations with frequencies equal to  $\sigma_m^k$ , unless of course the initial values of the normal mode coefficients  $y_m^k(0)$  were zero. This is strictly correct only for a continuous time dependence. The only effect of using the leap frog finite time difference approximation is, however, that the frequencies to be expected  $(\sigma_m^k)_c$  should be given by

$$(\sigma_m^k)_c = \frac{1}{\Delta t} \arcsin(\sigma_m^k \Delta t) \quad (12)$$

Actually time computational modes are also introduced, but due to the smooth start of the integration, mentioned in the introduction, the amplitudes of these should be extremely small compared to those of the physical modes. A closer inspection of the integration results showed that the amplitudes of the computational modes were indeed small during the whole integration period.

In order to facilitate a comparison between the theoretical frequencies and those found by the time series analysis the values of  $|(\sigma_m^k)_c|$  for three different basic states are shown by the solid lines in the right hand side of Figure 2a and in the two parts of Figure 2b, all of which otherwise are similar to the left hand side of Figure 2a.

The solid lines in the figures connect the points given by the values of  $|(\sigma_m^k)_c|$  for different zonal wave numbers so that if the theoretical frequencies were exactly equal to those found by the time series analysis then all the boxes should have their centres at the lines.

In the right hand side of Figure 2a is used a basic state at rest. It is seen from the figure that the observed frequencies fit the theoretical frequencies very well for low zonal wave numbers, whereas we find rather large discrepancies for the higher zonal wave numbers, especially for oscillations with small periods of time.

In the left hand side of Figure 2b is used a solid rotating basic state, with an angular velocity equal to 14.4 deg/day. It is seen in Figure 1, that this solid rotation roughly approximate the initial zonal velocity at medium and high latitudes. The theoretical frequencies for low zonal wave numbers are almost unchanged compared to those for a basic state at rest, however, we find a much improved correspondence between the observed and the theoretical frequencies for medium zonal wave numbers except for those oscillations with the smallest frequencies found for these wave numbers.

In the right hand side of Figure 2b is finally used a basic state determined by the initial zonal velocity field used in the numerical integration (cf. Figure 1). It is seen that with this basic state an almost complete correspondence is found between the observed and the theoretical frequencies.

It follows from Equations (7) and (9) that according to the linear theory the ratios between the amplitude of oscillations with a certain frequency  $\simeq (\sigma_m^k)_c$  found for different spherical harmonic coefficients  $x_m^j$  should be determined by the ratios between the components  $x_m^{jk}$  in the eigenvector  $X_m^k$ . Also in this respect we find a good correspondence between the linear theory and the results from the time series analysis.

We could have extended the comparison between the linear theory and the results of the time series analysis by comparing the normal mode coefficients  $y_m^k(0)$ , determined by the initial data, to estimated



values obtained from the time series analysis. We have done that only for gravity modes in the zonal mean field ( $m = 0$ ), and it is readily seen that for these components at least a discrepancy between the linear theory and the results found for the nonlinear model must be expected. As mentioned above the initial fields satisfy the linear balance equation, and as a consequence the initial normal mode coefficients  $y_m^k(0)$  with respect to a basic state at rest must be exactly zero for the gravity modes with  $m = 0$ . We should therefore not at all expect gravity oscillations in the zonal mean flow, at least not according to the linear theory based on a basic state at rest. Nevertheless, we do observe oscillations with frequencies almost exactly equal to those found by the linear theory and with ratios between the amplitudes which also correspond closely to those determined by the eigenvectors. This discrepancy can not be removed by using a more realistic basic state. Even with a basic state determined by the initial zonal mean velocity field, the values of  $y_m^k(0)$  for the gravity modes with  $m = 0$  become much too small to explain the observed oscillations.

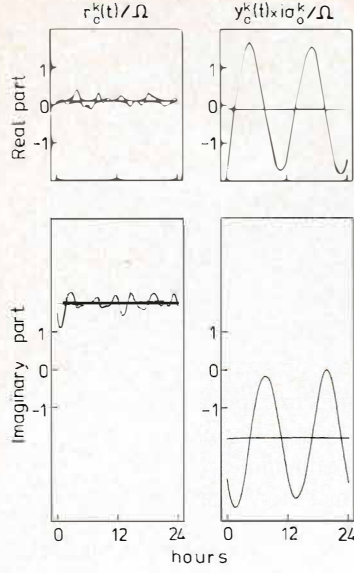
Thus, we see that the linear theory as presented in the preceding section fails to explain the amplitude of the gravity oscillations, at least those found in the zonal mean flow. However, the close correspondence in other respects, indicate that the gravity oscillations in the nonlinear model is determined to a high degree of accuracy by a time dependence of the normal mode coefficients, which are harmonic oscillations with frequencies equal to the eigenfrequencies  $(\sigma_m^i)_c$ , and with amplitudes influenced in some way or the other by the nonlinear interactions.

#### 4. Elimination of Gravity Oscillations Based on Normal Mode Solutions with Constant Nonlinear Forcing

In the preceding sections it was pointed out that in nonlinear integrations gravity oscillations still exist even though the normal mode coefficients of the gravity modes are reduced to zero initially. The reason for gravity oscillations still being present after the filtering must be that the linearizing assumption,  $r_m^k(t) \equiv 0$  in Equation (6), is not satisfied with sufficient accuracy for the gravity modes. In order to investigate this question and in order to see if a better approximation could be found we have, using "real" initial data, computed the nonlinear terms,  $r_m^k(t)$ , for the gravity modes in the present model during a 24 hours integration.

The initial data used were analysed height and wind fields from 550 mb, 4th November 1969 for the Northern Hemisphere, obtained from the global analysis of the GARP Basic Data Set, made by the Dynamic Prediction Research Division, Montreal. We shall refer to these data later as the BDS data. The global analysis have been distributed by the JOC Working Group on Numerical Experimentation without information about the procedure of analysis. As far as the author is informed, however, the analyses are based on the available observations only, without any use of forecast guess fields and without any special effort to obtain balanced data. We may therefore expect rather large amplitude gravity oscillations in forecasts based on these data.

The time series of the nonlinear terms,  $r_m^k(t)$ , for the gravity modes, using a basic state at rest, show that each  $r_m^k(t)$  is fluctuating around a slowly varying, and in many cases almost constant, mean value, which we shall denote  $R_m^k(t)$ . The value of  $R_m^k(t)$  is furthermore found to be significantly different from zero for many gravity modes. As an example is shown in Figure 3 the time variation of  $r_m^k/\Omega$  for the most large-scale gravity mode in the zonal mean field ( $m = 0$ ). For comparison is shown also the variation of  $i\sigma_m^k y_m^k/\Omega$ , the corresponding linear contribution to the tendency of  $y_m^k$  (cf. Equation (6)). The real parts are shown in the upper part and the imaginary parts in the lower part of the figure. For normal modes with  $m = 0$ , as the one chosen in the figure, the real parts contributes only to the tendency of the height and stream function fields and the imaginary parts contributes only to the tendency of the velocity potential field.



**Figure 3**

- Time variation of the linear contribution,  $\frac{1}{\Omega} i \sigma_m^k y_m^k(t)$  and the nonlinear contribution  $\frac{1}{\Omega} r_m^k(t)$  to the time derivative of the normal mode coefficient  $\frac{1}{\Omega} \frac{d}{dt} y_m^k$  for the most large-scale gravity mode with zonal wave number  $m = 0$ . The oscillating curves are from an integration with unfiltered initial data and the smooth curves from an integration with filtered initial data.

We may think of the smooth curves in the left hand side of Figure 3, around which  $r_0^k(t)/\Omega$  is oscillating, as curves for  $R_0^k(t)/\Omega$ . We shall later explain how these curves actually are determined.

Based upon the observed variation of  $r_m^k(t)$  for the gravity modes in the present numerical model we should expect to get an improved approximation to the solution of Equation (6) for these modes, by assuming each  $r_m^k(t)$  to be a constant, for which a reasonable choice would be the value of  $R_m^k(t)$  at  $t = 0$ . With this approximation we find the solutions

$$y_m^k(t) = -\frac{R_m^k(0)}{i\sigma_m^k} + \left( y_m^k(0) + \frac{R_m^k(0)}{i\sigma_m^k} \right) e^{i\sigma_m^k t} \quad (13)$$

Thus, we get harmonic oscillations around a constant mean value with frequencies equal to the eigenvalues  $\sigma_m^k$ . The solutions (13) differs from those given in Equation (7) when  $R_m^k(0) \neq 0$ . On the basis of these solutions we may now explain why we observe harmonic gravity oscillations in the integrations, even though the normal mode coefficients for the gravity modes have been reduced to zero initially. Especially it should be noted that all results of the time series analyses reported on in Section 3 can be explained on the basis of the solutions (13), including the gravity oscillations observed in the zonal mean flow.

Supposing Equation (13) to be valid for the gravity modes it is seen that the amplitude of the gravity oscillations can be reduced to zero by setting the normal mode coefficients  $y_m^k(t)$  at  $t = 0$  equal to corrected values  $(y_m^k(0))_c = -R_m^k(0)/(i\sigma_m^k)$ . From Equation (13) it is furthermore seen that

$$(y_m^k(0))_c = \frac{1}{T_m^k} \int_{-T_m^k/2}^{T_m^k/2} y_m^k(t) dt \quad (14)$$

where the period  $T_m^k = 2\pi/|\sigma_m^k|$ .

Thus, using periods corrected for the finite difference time scheme and an evaluation of the integral in (14) by a quadrature formula, the corrected values for the gravity modes might be evaluated from



the results of integrations forward and backward in time. We have made experiments with a filtering method building upon such a procedure and did obtain good results, but this method is rather expensive in terms of computing time, if we want to eliminate also the gravity oscillations for low zonal wave numbers with long periods. We have therefore given up this method and instead developed the method to be explained in the following.

By an iterative method we compute successive better estimates of  $R_m^k(0)$ . As a first guess we use the values  $(r_m^k(0))_0$ , that is, the values of  $r_m^k(t)$  at  $t = 0$  computed from the original data. First guess values of  $(y_m^k(0))_c$  are then determined by

$$(y_m^k(0))_1 = -\frac{(r_m^k(0))_0}{i\sigma_m^k} \quad (15)$$

which are substituted for the original values of  $y_m^k(0)$ , called  $(y_m^k(0))_0$ . The procedure may be repeated and it follows from Equation (9) that one iteration of the scheme is defined by

$$(x_m^j(0))_{\nu+1} = (x_m^j(0))_\nu + \sum_k ((y_m^k(0))_{\nu+1} - (y_m^k(0))_\nu) x_m^{jk} \quad (16)$$

where  $\nu$  is an iteration number and where the summation is carried out over only the values of  $k$  corresponding to the gravity modes. The values of  $(y_m^k(0))_{\nu+1}$  are determined by

$$(y_m^k(0))_{\nu+1} = -\frac{(r_m^k(0))_\nu}{i\sigma_m^k} \quad (17)$$

And the values of  $(y_m^k(0))_\nu$  and  $(r_m^k(0))_\nu$  are determined from the spherical harmonic coefficients  $(x_m^j(0))_\nu$  and  $(q_m^j(0))_\nu$  by solving the system of Equations (3) and (5). When using a basic state at rest, the eigenvectors become orthogonal and we get simply

$$\left. \begin{aligned} (y_m^k(0))_\nu &= \sum_{j=1}^N x_m^{jk} (x_m^j(0))_\nu \\ (r_m^k(0))_\nu &= \sum_{j=1}^N x_m^{jk} (q_m^j(0))_\nu \end{aligned} \right\} \quad (18)$$

For the present data set at least this iterative method seems to converge very fast. As we shall see in the next section the gravity oscillations in practice are eliminated completely after only two iterations.

The fast convergence may be explained as follows. It seems reasonable to ascribe the fast fluctuating part of  $r_m^k(t)$ , that is,  $r_m^k(t) - R_m^k(t)$ , to the nonlinear interactions between the oscillating part of the gravity modes. Already the substitution of  $(y_m^k(0))_1$  for the original values  $(y_m^k(0))_0$  results in a substantial reduction of the amplitudes of the gravity oscillations, and as a consequence the fast varying part of  $r_m^k(t)$  is also reduced very much. Therefore  $(r_m^k(0))_1$  becomes a good approximation to  $R_m^k(0)$ .

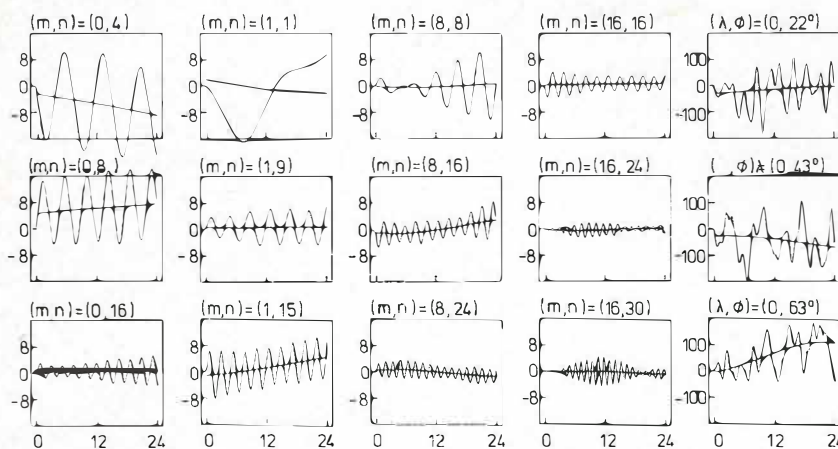
Referring again to Figure 3, the smooth curves describe the variation of the linear and the nonlinear contributions to the tendency computed from initial data filtered by the procedure described above, with  $\omega = 0$  and using two iterations. It is seen that the sum of the linear and the nonlinear contribution to the tendency for this normal mode coefficient is close to zero during the whole integration period so that the coefficient itself remains almost constant.

## 5. Results of the Filtering Experiments

The filtering procedure described in the preceding section, using a basic state at rest and two iterations, was applied to the BDS data from 4th November 1969. In order to illustrate the performance of this new procedure we have in Figure 4 shown the time variation of twelve cosine spherical harmonic height components and the height at three points at the Greenwich meridian during two 24 hours integrations. The oscillating curves show the results of an integration with the unfiltered initial data and the smooth curves show the results with filtered initial data. In the figure the deviations from the initial values of the unfiltered fields are given in metres. For each cosine component is shown the value of the root mean square deviation (computed as the square root of the contribution to the mean square deviation) times the sign of the deviation. To judge from the figure the gravity oscillations have been completely eliminated by the filtering procedure, and furthermore once these oscillations have been eliminated, they do not seem to redevelop. These conclusions are confirmed by a closer inspection of the integration results.

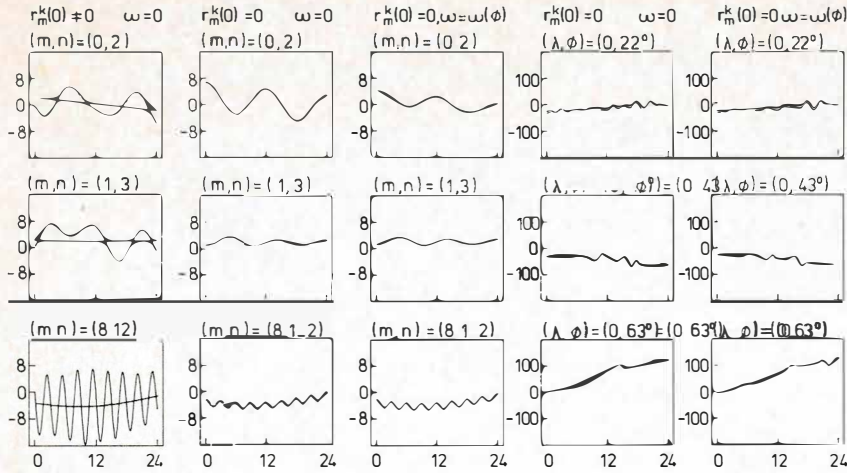
The filtering procedure results in substantial changes of the height field as well as of the divergent velocity field, whereas the nondivergent velocity field is changed rather little. The root mean square deviations between the filtered and the unfiltered initial fields are 6.2 m, 2.9 m/s and 0.6 m/s respectively. After the filtering the velocity field has become almost nondivergent, the divergent velocity field being weak and very large scale, and the patterns of the height and streamfunction has become quite similar, indicating approximately geostrophic or gradient wind balance at high and medium latitudes. We have, however, not investigated to what extent such a balance exist.

Curve No. 1 in Figure 6 shows the root mean square deviation (RMSD) between the two integrations for the height field. It is seen that the value oscillates around a constant mean value during the whole five days period. This turns out to be the case also for the RMSD's of the divergent and the non-divergent velocity field. These facts indicate that the gravity oscillations present in the integration with unfiltered initial data neither grow nor influence significantly the slowly varying part of the fields. This indication is confirmed as it turns out, that we get almost the same result if we apply the filtering procedure to the forecast instead of applying it to the initial data. We have computed the

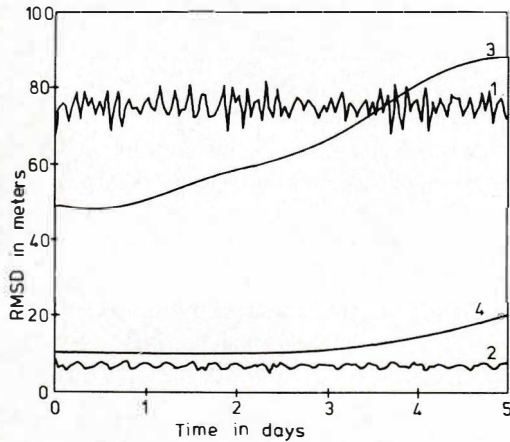


● **Figure 4.** Time variation in filtered and unfiltered forecasts of twelve cosine spherical harmonic height coefficients and the height in three points at the Greenwich meridian in metres.





● **Figure 5.** Time variation of three cosine spherical harmonic components and the height in three points at the Greenwich meridian, with different initial data, determined as indicated above each column. In the first column from the left is shown also the time variation obtained with unfiltered initial data.



● **Figure 6**

RMSD of  $h-H$  between an integration with given initial data = I.D. and an integration with initial data determined by the new normal mode filtering procedure. Curve No. 1 (No. 2) is for I.D. equal to the unfiltered data (data obtained by free normal mode filtering). Curve No. 3 (No. 4) is for I.D. obtained by forced adjustment toward the unfiltered height field (the unfiltered stream function field).

RMSD's between the filtered 24 hours forecast with unfiltered initial data and the 24 hours forecast with filtered initial data, and find the values 0.4 m, 0.23 m/s and 0.01 m/s for the height field, the divergent and nondivergent velocity field, respectively. These observations may be explained as follows. In forecasts with unfiltered initial data we find, also for the normal mode coefficients of the Rossby type, that the nonlinear contribution to the tendency,  $r_m^k(t)$ , may be split up in two parts, a slowly varying part,  $R_m^k(t)$ , and a fast fluctuating part,  $r_m^k(t) - R_m^k(t)$ . Here the latter part is the deviation from the nonlinear tendency in a forecast with filtered initial data and therefore this part is due to interactions involving the oscillating part of the gravity modes. At a certain time this fluctuating part may be significantly different from zero, but over a sufficient long period of time the integrated effect of this part of the tendency is very small.

In Figure 5 is shown some results from two integrations where we have used the filtering method proposed by DICKINSON and WILLIAMSON (loc. cit.), presented in Section 2 for the present model. In both integrations the normal mode coefficients for all gravity modes were reduced to zero initially, but two different basic states were used.

We shall at first look at the results from one of the integrations in which the filtering was based on a basic state at rest. In the fourth column is shown the time variation of the height in the three points at the Greenwich meridian, considered also in Figure 4. It is seen that this "free normal mode filtering" results in a considerable reduction of the fluctuations due to gravity oscillations. Some fluctuations are, however, still observed after the filtering, and it is seen that fast changes up to about 50 m still occur. In the second column is shown the time variation for three cosine spherical harmonic components. For comparison are shown in the first column the results for the same three components with unfiltered initial data and with initial data filtered as described in Section 4 taking into consideration the nonlinear forcing. It is seen that for these components a substantial improved filtering is obtained by taking into consideration the nonlinear forcing. The three components have been chosen in order to show that large differences may occur between the two methods. Generally such big differences are found only for rather large scale components. For small scale components the free normal mode filtering almost completely eliminate the gravity oscillations, and the total RMSD's between integrations with initial data filtered by the two methods are rather small. Curve No. 2 in Figure 6 shows the RMSD for the height field, and it is seen that the value fluctuates around a value of approximately 8 m.

In order to see to what extent an improvement could be achieved in the free normal mode filtering by using a more realistic basic state than a state at rest, we made another integration from which some results are shown in Figure 5; this time using a basic state defined by the nondivergent zonal velocity field determined by the unfiltered initial data. It should be noted that the procedure in this case is more complicated than in the case with a basic state at rest. As the eigenvectors are not orthogonal the normal mode coefficients can not be determined by Equation (11), but must be computed by solving Equation (3). In the present case the solution of this equation became further complicated by the fact that the eigenfrequencies  $\sigma_m^k$  and the eigenvectors  $X_{m1}^k$  for some of the Rossby modes with low zonal wave numbers became complex due to barotropic instability of the basic state. Although the instability was very weak, with e-folding times of about 25 days, the imaginary parts of  $X_{m1}^k$  had to be taken into account. Some integration results with initial data computed by this filtering method is shown in column No. 3 and 5 of Figure 5. It is seen that only small changes compared with the results obtained using a basic state at rest have been achieved. A conclusion which is confirmed by a closer inspection of the integration results.

Summarizing the above results we find that the new filtering method introduced in Section 4 imply a significant improvement for the gravity oscillations with rather large scales (low frequencies), whereas the nonlinear interactions have only small effects on the filtering of small-scale (high-frequency) oscillations. The reason for this fact must be that for high-frequency modes the nonlinear term,  $r_m^k$ , become small compared to the linear term  $i\sigma_m^k y_m^k$  in Equation (6). Concerning the importance of the choice of the basic state we find that although a better agreement with the frequencies observed in the nonlinear model is achieved when using a realistic basic state, the results show that it is sufficient to use a basic state at rest in the filtering procedure.

## 6. The Application in Initialization Procedures of the New Filtering Method

An extension of the filtering method, described in Section 4 for the barotropic model, to more realistic baroclinic models implies a determination of the normal modes for such models. A



general approach to determine the normal modes of a specific numerical prediction model was given in DICKINSON and WILLIAMSON (1972) and illustrated by application to a global finite difference two-layer model. Recently the normal modes of the NCAR global circulation model (GCM) linearized about a basic state at rest were determined by WILLIAMSON and DICKINSON (1976).

For a multi-layer model a separation of variables approach is used, that is, the normal modes are assumed to be products of vertical and horizontal modes. The vertical modes are given by vertical structure equations, determining discrete series of equivalent depths,  $H_l$ , and corresponding vertical modes, equal in number to the number of layers in the model. The vertical modes are determined at the discrete levels and may be numbered in the order of decreasing equivalent depth. The first vertical mode, the external mode, is of the same sign at all levels, and almost a constant. For the other modes, the internal modes, the number of zero crossings are increasing by one for each smaller equivalent depth. For each equivalent depth,  $\bar{H}_l$ , the horizontal modes are equal to those of the linearized shallow water model, with a mean height equal to  $\bar{H}_l$ . The frequencies of corresponding gravity modes with different equivalent depths,  $\bar{H}_l$ , decrease with decreasing  $\bar{H}_l$  as indicated by the decrease of the speed of pure gravity waves,  $(g\bar{H}_l)^{1/2}$ . The mean height in the shallow water model used in the foregoing sections ( $\bar{H} = 8800$  m) corresponds approximately to the equivalent height for external modes found in baroclinic models ( $\approx 10^4$  m).

The normal modes form a complete set, so that it is always possible to expand the model data, in terms of these modes. It seems likely that the gravity modes, at least those corresponding to the largest equivalent depths and therefore those with high frequencies, also for a baroclinic model behave nearly as predicted by the linear theory. A free normal mode filtering of unwanted gravity modes should therefore be possible also for baroclinic models, and we should expect such a filtering to be improved by taking into account the nonlinear interactions, as described in the preceding sections. We have made a preliminary trial in this direction, which indicate that at least external gravity oscillations can be eliminated by this method. The model used was an inviscid and adiabatic spectral five-level model, similar to that described by HOSKINS and SIMMONS (1975), and the method used was the time averaging method, described in connection with Equation (14), applied to the vertical mean values of the variables of the model. The vertical mean is, however, only approximately equal to the external mode of the five level model and furthermore the time averaging method is very expensive in terms of computing time, if we want to eliminate the gravity oscillations with rather long periods. Therefore, a more ideal filtering procedure should imply an expansion in terms of the vertical modes and for each vertical mode the application of the iterative method as described in Section 4.

In the foregoing we have considered methods by which gravity oscillations may be eliminated from the initial state. Whether a given oscillation should be eliminated at all, must be a question of whether the oscillation in the model simulates a real phenomenon in the atmosphere; and if so, if the normal mode coefficient can be determined by the meteorological observing system with sufficient accuracy.

At the present time these questions are very difficult to answer definitively. The general opinion is, however, that large scale high frequency gravity oscillations are at least of very small amplitude in the atmosphere, and that we cannot determine these small amplitudes with sufficient accuracy. Therefore the atmosphere must be close to a balanced state, characterized by being a state without such gravity waves.

We may obtain such a balanced state, consistent with a given numerical model, by filtering out the high frequency gravity oscillations from a given data set. When using the filtering procedure, defined in Section 4 for the shallow water model, the balanced state is defined as a state for which the time derivatives of the normal mode coefficients for the gravity modes are zero. Thus, the balanced state is defined by the filtering procedure and here we should of course only filter out those oscillations,

which have a small amplitude in the atmosphere. With this definition of a balanced state, the filtering procedure results in an instantaneous adjustment to a balanced state, and we shall therefore in the following refer to it as the adjustment procedure.

Concerning the application of the adjustment procedure in an initialization method, we may think of applying the procedure to some data (obtained by some objective analysis scheme), which are fitting the observations within certain limits. (Some small deviations must be tolerated because of observation errors and small scale effects). For the shallow water model we saw in the preceding section, that the adjustment procedure, at least when applied to the BDS data, results in large changes of the data. If these results can be considered representative for the external mode in a more realistic model, it follows that the adjusted fields no longer fit the observations well, especially because the adjustment procedure results in large changes in the height field. Therefore the mere adjustment of some analysed data does not seem to be an ideal initialization procedure.

In order to estimate the order of magnitude of the changes resulting from the adjustment procedure in a more realistic model and applied to data obtained by an advanced objective analysis scheme, we may refer to some results of SCHLATTER *et. al.* (1976). They have developed a statistical interpolation scheme with a geostrophic constraint in a medium and high latitudes and a nondivergent constraint combined with a balance equation in the tropics. (The constraints are only weak; allowing ageostrophic and divergent winds in the final analysis). Furthermore “first guess” fields from 12 hour forecasts with the NCAR GCM are used. With data obtained by this analysis scheme and applied to the NCAR GCM, without any filtering of the data, they find at a grid point over USA short-period oscillations in the surface pressure, with amplitude occasionally greater than 10 mb (see their Figure 17). These oscillations are found especially during the first part of the integration period. The mean value around which the pressure seems to oscillate differs initially from the actual initial pressure by as much as the amplitude of the oscillations, a fact which indicates that we would get pressure changes of the order of magnitude of 10 mb in data-rich areas as a result of our adjustment procedure. For comparison the RMSD between analysed and observation station values of geopotential height near the surface are found to be of the order 8 m, corresponding to 1 mb. (See their Figure 10).

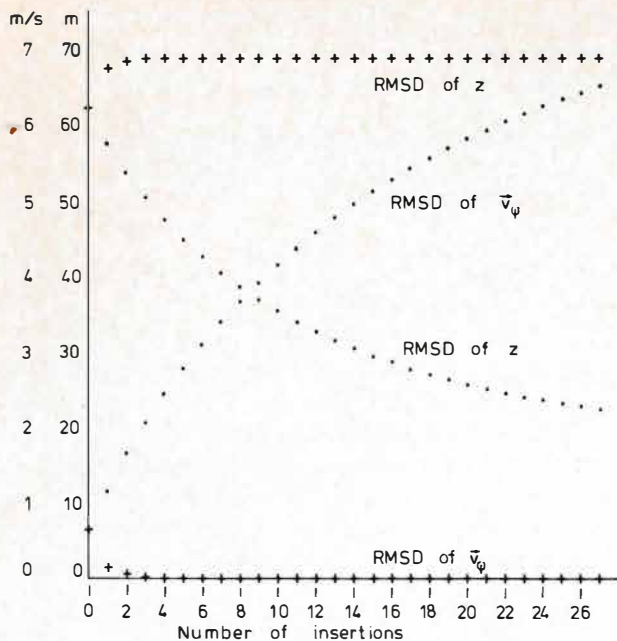
Of course the above estimates of changes as a result of the adjustment procedure should be taken with some reservation. If, however, the estimates are right, then it seems possible to improve a procedure, consisting only of an analysis followed by an adjustment procedure.

To our opinion the “best” initialization scheme determine a balanced state, which fit the observations as well as possible, and where the balanced state is defined as stated above. The most ideal scheme may be an integrated analysis – adjustment scheme, which incorporates as a strong constraint, this balance condition; namely, that the time derivative of the normal mode coefficients should be zero for those coefficients corresponding to upward oscillations.

We have not considered in detail if such a system can be developed and shall instead suggest the following iterative procedure: After adjusting the original analysed fields the analysis scheme should be repeated, using the balanced fields as the “first guess” field. The procedure consisting of an analysis and an adjustment step should then be repeated until no more reduction can be achieved of the difference between the balanced fields and the observations at the observation points. We think that such a procedure will converge and that it will result in significant changes in the forecasts as compared to those obtained with the original balanced fields as initial data. The following experiments indicate that this is the case.

Again the experiments were carried out with the shallow water model, using the BDS data. In the first experiment we assumed that the original height field was “correct” and in the second one that the original stream function field was “correct”. The adjustment procedure (carried out as described in Section 4, using a basic state at rest and two iterations) was repeated and after each adjustment





● Figure 7

RMSD's between the original, unbalanced fields and the balanced fields, obtained by forced adjustment, as functions of the number of insertions of the original height field (indicated by dots) and the original stream function field (indicated by crosses).

procedure the field assumed to be "correct" was restored. The resulting RMSD's between the original unbalanced fields and the balanced fields obtained by the procedure are shown in Figure 7 as functions of the number of insertions. It is seen that the procedure converge; in the case of restoring the height field rather slowly, and in the case when the stream function field is restored rather fast. The effect on forecasts of these forced adjustment procedures applied to the initial data are indicated in Figure 6. The curve No. 3 (No. 4) shows for the height field the RMSD between two intergrations, namely one with initial data equal to the original balanced fields and one with initial data equal to those obtained after 26 insertions of the height field (the stream function field). It is seen that in both cases the RMSD is increasing during the integration period, indicating that significant changes have taken place during the forced adjustment processes.

It should be emphasized that the balanced state to which the forced adjustment process approaches is very much different from a state where the gravity normal mode coefficients are zero. So, we obtain very large amplitude gravity oscillations when using an initial state determined by expanding the the original height field in terms of the Rossby modes (that is, the original height field is represented exactly by the Rossby modes and all gravity mode coefficients are zero). The amplitudes of the gravity oscillations are even larger in this case than when using the original, unmodified data fields as initial data in the model.

It is interesting to note the difference between the curves No. 1 and 2 on the one hand and the curves No. 3 and 4 on the other hand, which show that only changes in the initial state of the Rossby modes result in significant changes of the forecast. The finding in the preceding section, that we get almost the same result whether we adjust the initial fields or the 24 hour forecast, is in agreement with results obtained by WILLIAMSON (1976), when using the free normal mode filtering procedure. It should be pointed out, however, that his as well as our results were obtained for the shallow water model; for a more realistic integration model, including precipitation processes, it is likely that large

amplitude gravity oscillations may significantly influence the precipitation amount and distribution. Thereby they may indirectly influence the meteorologically significant part of the motion and mass fields. (c.f. THEMPERTON (1973) for a discussion of this subject). As explained above we think, furthermore, that it is not sufficient just to eliminate the gravity oscillations initially; gravity oscillations are sign of errors in the initial data and a forced adjustment toward the observed winds and heights seems to be necessary, in order to get a balanced state which fits the observations sufficiently well.

## 7. Conclusions

The gravity oscillations in a shallow water model, with a mean height approximately equal to the equivalent height for external modes in baroclinic models are studied. For this model the dynamics of the gravity oscillations are found to be governed, with a high degree of precision, by quasi-linearized equations, where the nonlinear terms are treated as constants in time. A balanced state may be defined as a state where the time derivative of the normal mode coefficients for the gravity modes are equal to zero. The balanced state corresponding to a given unbalanced state may be determined by an iterative procedure, called the adjustment procedure. The balanced state seems to be determined with sufficient accuracy using gravity modes based on a basic state at rest. In the model the nonlinear contributions,  $r_m^k(t)$ , to the time derivative of the normal mode coefficients  $\frac{d}{dt}y_m^k$ , are fluctuating around slowly varying values,  $R_m^k(t)$ , equal to  $r_m^k(t)$  in the corresponding balanced state. In the quasi-linearized equations the nonlinear terms  $r_m^k(t)$  are approximated by the corresponding  $R_m^k(0)$ , defined by that balanced state which is obtained from the initial data. The inclusion of the nonlinear terms,  $R_m^k(0)$ , is important especially for relatively large-scale (low-frequency) gravity modes.

In agreement with the quasi-linear approximation, leading to the solutions given in Equation (13), the gravity oscillations which are generated with unbalanced initial data appear neither to grow nor to decline during the integration period and no gravity oscillations seems to develop, when using a balanced initial state. Furthermore, when present the gravity oscillations seem not to influence the balanced state, significantly.

It seems likely that the above results may be extended to baroclinic models, at least to the external modes in such models.

The experiments carried out with unbalanced real data using the present model, and an estimate based on results with the NCAR GCM indicate that the adjustment procedure can lead to rather large changes in the initial fields. As a consequence a forced adjustment procedure (or the incorporation of the balance conditions in the analysis scheme), may be necessary in order to determine a balanced state, which fits the observations sufficiently well. Experiments with the shallow water model indicate, that a simple forced adjustment procedure, which reduces the deviation between the resulting balanced initial fields and the observations, can be developed and that changes in the initial state resulting from such a procedure may result in significant changes of the subsequent forecast.

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